DIVISIBILITY OF THE FACTORIAL RATIO $(8n)!\,n!/\big((4n)!\,(3n)!\,(2n)!\big)$: PROOFS OF CONJECTURES OF BALA

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Abstract. Let $a(n) = (8n)!\,n!/((4n)!\,(3n)!\,(2n)!)$ (OEIS A211420). Peter Bala conjectured that $(n+1) \mid 35\,a(n)$, $(2n+1) \mid 3\,a(n)$, $(3n+1) \mid 5\,a(n)$ and $(8n-1) \mid a(n)$ for all n. He also conjectured that the products $(kn+1)\cdots(kn+r)$ for $k \in \{1,2,3\}$ and $(8n-1)(8n-3)\cdots(8n-2r-1)$ divide fixed multiples of $a(n)$. We prove all of these. The divisibilities $(n+1) \mid 35a(n)$, $(2n+1) \mid 3a(n)$ and $(8n-1) \mid a(n)$ appear to be new, and the constants 35, 3 and 5 are optimal. The proofs evaluate Legendre's formula through a single step function of period 24. At the primes 2 and 3 we use a short "escape" lemma. All four statements, as formalized in Google DeepMind's Formal Conjectures repository, are proved in Lean 4 with Mathlib.

1. Introduction

The numbers $a(n) = \frac{(8n)!\,n!}{(4n)!\,(3n)!\,(2n)!} = 1, 140, 60060, 29745716, \ldots$ form OEIS A211420. They are integers by Landau's criterion; they also belong to a family studied by Bober [2]. In 2025 Bala recorded the following conjectures in the OEIS entry [5]:

(C1) $(n+1) \mid 35\,a(n)$, $(2n+1) \mid 3\,a(n)$ and $(3n+1) \mid 5\,a(n)$ for all $n \geq 0$;
(C2) for $k \in \{1,2,3\}$ and $r \geq 1$ there is $C(k,r) > 0$ with $(kn+1)(kn+2)\cdots(kn+r) \mid C(k,r)\,a(n)$ for all n;
(C3) $(8n-1) \mid a(n)$ for all n;
(C4) for $r \geq 0$ there is $K(r) > 0$ with $\prod_{i=0}^{r}(8n-2i-1) \mid K(r)\,a(n)$ for all n.

They are stated in the Formal Conjectures repository [4] as `OeisA211420.conjecture1`, …, `conjecture4`.

Theorem 1. *(C1)–(C4) hold. In (C1) the constants $35, 3, 5$ cannot be decreased. One can take $C(k,r) = ((24r)!)^r$ and $K(r) = ((6r+3)!)^{r+1}$.*

Prior work. The divisibility $(3n+1) \mid 5a(n)$ is the case $k=4$ of Sun's theorem $\binom{2n}{n} \mid (k+1)'\,C_n^{(k-1)}\binom{2kn}{kn}$ [6], because $a(n) = \binom{8n}{4n}\binom{4n}{n}/\binom{2n}{n}$. Yang [7] proved $(2n+1)(2n+3) \mid 99\,a(n)$. Statement (C2), with the explicit constant $\operatorname{lcm}(1,\ldots,8r-1)^r$, was proved and formalized in Lean in [3]. Statement (C4), with $K(r) = (8r)!$, was proved by Adamczewski [1, Appendix A]. For $r=0$ this gives only $(8n-1) \mid 8!\,a(n)$. We found no earlier proof of $(n+1) \mid 35a(n)$, $(2n+1) \mid 3a(n)$ or $(8n-1) \mid a(n)$.

2. Legendre's formula as a residue function

For a prime power $q = p^j$ put $T = \lfloor 24n/q \rfloor$. For $c \in \{1,2,3,4,8\}$ we have $c \mid 24$, and therefore $\lfloor cn/q \rfloor = \lfloor T/(24/c) \rfloor$. So Legendre's formula gives

$$v_p(a(n)) = \sum_{j \geq 1} \varphi\big(\lfloor 24n/p^j \rfloor\big), \qquad \varphi(T) = \left\lfloor \frac{T}{3} \right\rfloor + \left\lfloor \frac{T}{24} \right\rfloor - \left\lfloor \frac{T}{6} \right\rfloor - \left\lfloor \frac{T}{8} \right\rfloor - \left\lfloor \frac{T}{12} \right\rfloor. \tag{1}$$

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Since $8+1-4-3-2=0$, φ has period 24. By inspection of $T=0,\dots,23$,

$$\varphi(T)=1 \iff T \bmod 24 \in G := \{3,4,5,6,7,9,10,11,15,21,22,23\}, \tag{2}$$

and $\varphi(T)=0$ otherwise. In particular $\varphi \geq 0$, which is the integrality of $a(n)$. To prove $t \mid c\,a(n)$ it suffices, for every prime p and every $j \geq 1$ with $p^j \mid t$, that $p^j \mid c$ or $\varphi(\lfloor 24n/p^j \rfloor)=1$. We call this the *termwise* criterion. Write $T_j = \lfloor 24n/p^j \rfloor$, so that $T_{j+1} = \lfloor T_j/p \rfloor$.

Lemma 2 (location)**.** *Let $q \geq 2$ and let m denote the cofactor in each case.*

(L1) *If $q \mid n+1$ and $q \geq 8$, then $T \bmod 24 \in \{21,22,23\}$.*
(L2) *If $q \mid 2n+1$ and $q \geq 4$, then $T \bmod 24 \in \{9,10,11\}$.*
(L3) *If $q \mid 3n+1$, $q \geq 4$ and $q \neq 5$, then $T \bmod 24 \in \{6,7,15,23\}$.*
(L4) *If $q \mid 8n-1$ and $q \geq 4$, then $T = 3m$ with m odd, so $T \bmod 24 \in \{3,9,15,21\}$.*

In all four cases $\varphi(T)=1$.

Proof. (L1): $n+1=qm$ gives $24n = 24mq-24$ with $24 \leq 3q$, so $24m-3 \leq T \leq 24m-1$. (L2): m is odd and $24n = 12mq-12$ with $12 \leq 3q$, so $12m-3 \leq T \leq 12m-1$. (L4): $8n-1=qm$ gives $24n = 3mq+3$ with $3<q$, so $T=3m$. (L3): for $q \geq 8$ we get $T=8m-1$, whose residue is 7, 15 or 23. For $q=4$, $n \equiv 1 \pmod 4$ gives $T = 6n \equiv 6$. For $q=7$, $n \equiv 2 \pmod 7$ gives $T \equiv 6$. The case $q=6$ cannot occur. □

Lemma 3 (escape)**.** *Let $p=3$ and suppose $T_i \bmod 24 \geq 9$; or let $p=2$ and suppose $T_i \bmod 24 \geq 6$. Then $\varphi(T_j)=1$ for some $j>i$.*

Proof. Let $p=3$ and suppose $\varphi(T_j)=0$ for all $j>i$. We show by downward induction that $T_j \bmod 24 \leq 8$ for all $j \geq i$; the start is $T_j=0$ for large j. If $S = T_{j+1}$ has $\varphi(S)=0$ and $S \bmod 24 \leq 8$, then $S \bmod 24 \in \{0,1,2,8\}$ by (2), so $S \bmod 8 \leq 2$. Hence $T_j \bmod 24 = 3(S \bmod 8) + (T_j \bmod 3) \leq 8$. This contradicts $T_i \bmod 24 \geq 9$. For $p=2$ the same argument works with $S \bmod 24 \in \{0,1,2\}$ and $T_j \bmod 24 = 2(S \bmod 12) + (T_j \bmod 2) \leq 5$. □

3. Proofs

Proof of (C3). At $n=0$ the divisor is -1. Let $n \geq 1$; then $t = 8n-1$ is odd. For $p \geq 5$ every p^j is at least 5, so the termwise criterion holds by (L4). For $p=3$ let $e = v_3(t) \geq 1$ and $t = 3^e m$. The indices $j=2,\dots,e$ contribute $e-1$ by (L4). Moreover $24n = 3^{e+1}m+3$ gives $T_{e+1}=m$. From $3^e m \equiv 7 \pmod 8$ we get $m \equiv 5$ or $7 \pmod 8$, and $3 \nmid m$. So $m \bmod 24 \in \{5,7,13,23\}$. Either $\varphi(m)=1$, or $m \equiv 13$, in which case Lemma 3 gives an index $j > e+1$ with $\varphi(T_j)=1$. Hence $v_3(a(n)) \geq e$. □

Proof of (C1). $(2n+1) \mid 3a(n)$. Here 2 does not divide $2n+1$, so let $q = p^j \mid 2n+1$ with p odd. If $q \geq 4$, (L2) applies; if $q=3$, then $q \mid 3$.

$(3n+1) \mid 5a(n)$. For odd p the criterion holds termwise by (L3), since $3 \nmid 3n+1$ and $5 \mid 5$. For $p=2$ let $e = v_2(3n+1)$ and $3n+1 = 2^e u$. The indices $j=2,\dots,e$ contribute $e-1$ by (L3). If $e=1$, then $n \equiv 3 \pmod 4$ and $T_2 = 6n \equiv 18 \pmod{24}$. If $e \geq 2$, then $T_{e+1} = 4u-1$, whose residue is 3, 11 or 19. In each case $\varphi(T_{e+1})=1$, or $T_{e+1} \bmod 24 \geq 6$ and Lemma 3 supplies the missing index.

$(n+1) \mid 35a(n)$. For $p \notin \{2,3\}$ the criterion holds termwise: by (L1) if $q \geq 8$, while $q<8$ forces $q = p \in \{5,7\}$, and $p \mid 35$. For $p=3$ let $e = v_3(n+1)$ and $n+1 = 3^e m$. The indices $j=2,\dots,e$ contribute $e-1$ by (L1). Also $T_{e+1} \in [8m-3, 8m-1]$, with residue in $\{5,6,7,13,14,15,21,22,23\}$. So $\varphi(T_{e+1})=1$, or the residue is 13 or 14 and Lemma 3 applies. For $p=2$ let $e = v_2(n+1)$ and $n+1 = 2^e u$. The indices $j=3,\dots,e$ contribute $e-2$ by (L1). The value T_{e+1} equals $12u-6$, $12u-3$, or lies in $[12u-2, 12u-1]$, according as $e=1$, $e=2$ or

$e \geq 3$. Its residue is 6, 9, 10 or 11, so $\varphi(T_{e+1}) = 1$ and $T_{e+1} \bmod 24 \geq 6$. Lemma 3 then gives a further index, for a total of at least e.

Optimality. $7 \mid n+1$ but $a(6) \equiv 1 \pmod 7$; $5 \mid n+1$ but $a(9) \equiv 1 \pmod 5$; $a(1) = 140 \not\equiv 0 \pmod 3$; and $3n+1 = 10$ while $a(3) = 29745716 \not\equiv 0 \pmod 5$. □

Proof of (C2) and (C4). For (C2), let $P = (kn+1)\cdots(kn+r)$. Then $v_p(P) = \sum_j c_j$, where c_j is the number of multiples of $q = p^j$ in $(kn, kn+r]$. If $q \leq 24r$, then $c_j \leq r \leq r\lfloor 24r/q \rfloor$, which is covered by $v_p(((24r)!)^r)$. If $q > 24r$, then $c_j \leq 1$. When $c_j = 1$ the multiple is $kn+i$ with $24i < q$; writing $D = 24/k$, we get $T_j = Dm - 1$ and hence $\varphi(T_j) = 1$.

For (C4), let $\ell_i = |8n - 2i - 1|$, which is odd and nonzero. Prime powers $q \leq 6r+3$ are covered by $v_p(((6r+3)!)^{r+1})$. If $q > 6r+3$ and $q \mid \ell_i$, then $8n > 2i+1$ and $8n = qm + 2i + 1$ with $3(2i+1) < q$. So $T_j = 3m$ with m odd and $\varphi(T_j) = 1$. Since m determines i, at most one ℓ_i is divisible by q. □

4. Formalization and checks

All four statements of [4] are proved in Lean 4 with Mathlib, against the unmodified statement file. The theorems `OeisA211420.conjecture1_solved`, …, `conjecture4_solved` have the types of the repository statements; this is checked by `rfl`. They depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The formal proofs follow Sections 2–3, with Legendre's formula taken from Mathlib. We also checked (C1) and (C3) numerically for all $n \leq 200\,000$ via (1), and with exact integers for $n < 400$. (C2) and (C4) were checked for $r \leq 6$ and $n \leq 3000$.

References

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